YUANJUN (JOHNNY) DAI

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SUMMARY

AI Infrastructure Engineer and PhD Candidate at CWRU, specializing in distributed ML training systems, LLM deployment, and eBPF-based observability. 8 research papers (7 first-author) at IEEE, ACM, and Springer; 5+ years of industry experience at **Cisco**, **Broadcom**, **Midea**, and **Xiaomi**.

SKILLS

Languages	Python, C/C++, Java, Bash
AI / ML	PyTorch (DDP/FSDP), TensorFlow, BytePS, Megatron-LM, NCCL, DeepSpeed, vLLM, TensorRT-LLM
Infrastructure	Kubernetes, Docker, CI/CD, ArgoCD, Helm, Slurm, HPC, GitHub Actions
Observability	eBPF, Prometheus, Grafana, DCGM Exporter, Intel RAPL, Kernel Tracing
Cloud & HPC	Lambda Labs HPC, NSF FABRIC, CloudLab, SDN (OpenFlow / Ryu), P4, NPU/ASIC (Broadcom BCM SDK)
Research	Distributed ML Systems, ML Security, Side-Channel Analysis, RL-based Scheduling

INDUSTRY EXPERIENCE

Software Engineer II — 06/2015 – 08/2016
Cisco Systems, Inc.

- Developed NPU/ASIC features for a 100 Gbps core router (Azure, AWS, Alibaba Cloud) using Broadcom BCM SDK.
- Contributed to QoS and ECMP modules, reducing forwarding latency under heavy load.

Applied Research Engineer — 09/2016 – 06/2018
Midea Group

- Led ML model training and deployment for **Fun Oven II**; designed data pipeline, applied transfer learning, and deployed Docker-based services on Alibaba Cloud.
- Product debuted at **2018 AWE Show** and won the **Red Dot Design Award** · Red Dot ↗

Technical Product Manager — 07/2018 – 01/2019
Xiaomi

- Defined system-level architecture for an IoT-enabled smart appliance; translated product constraints into deployable system designs and aligned engineering and business stakeholders.

Software Engineer Intern — 10/2014 – 05/2015
Broadcom (Emulex)

- Replaced polling-based data acquisition with a DMA + circular-buffer architecture in an RTOS pipeline, **reducing CPU load by 30%** and improving sampling rate and control-loop responsiveness.
- Ported a Linux-only management CLI for BladeEngine ASIC-based CNA hardware to Windows Server.

EDUCATION

Ph.D. Candidate in Computer Science (AI Infrastructure) — 08/2019 – Sep. 2026 (exp.)
Case Western Reserve University

M.S. in Electrical and Computer Engineering (Computer Microarchitecture)
University of Pittsburgh

B.S. in Electrical Engineering
Northeastern University

RESEARCH & SYSTEMS ENGINEERING EXPERIENCE

Research Assistant, AI Infrastructure & ML Systems
Case Western Reserve University

- **DYNAMIX – RL-Based Adaptive Batch Scheduling for Distributed ML** | Paper / Code
  PPO-based centralized RL scheduler dynamically adapting batch size via real-time system signals; evaluated on PS (BytePS) and Ring AllReduce (PyTorch DDP) architectures; achieves **46% training speedup** over optimal fixed batch size strategy with improved model generalization, on up to **64 A100 GPUs** at Ohio Supercomputer Center and Lambda Labs HPC.
- **PRISM – Priority-Aware Transport for Distributed ML Training** | Paper (Under Review)
  Priority-based selective reliability transport between NCCL/Gloo and UDP; magnitude-based gradient classification with full retransmission for high-priority and local compensation for low-priority; dual congestion windows, adaptive window borrowing,

GPU-accelerated TopK, SACK-based tracking, kernel-bypass (LibVMA) and GSO/GRO optimizations; achieves **8–15% throughput gain for CNNs and 15–21% for LLMs** over TCP on 32 GPUs (OSC) and cross-site FABRIC testbed; matches NDP latency at 144-worker scale on commodity infrastructure.

- **Training-Time Side-Channel Attacks on Distributed ML Systems** | Paper / Code
  Analyzed MXNet BSP PS-architecture communication patterns via push/pull instrumentation; modified MXNet source code for ground-truth collection (framework instrumentation); cross-correlated network traffic and Intel RAPL power side channels with outlier filtering and polynomial regression for energy profiling; achieved **>99% layer-type accuracy and <1.5% tensor-size error** across ZFNet, AlexNet, and VGG variants — exposing DNN architecture stealing surfaces in MLaaS co-location environments.
- **Priority-Based eBPF Network Flow Telemetry for AI / DML Infrastructure** | Paper / Code
  In-kernel eBPF priority-aware sketch framework (priority table + heavy flow table + 3-layer CMS); sequence-number-based retransmission detection; atomic operations for thread-safe concurrent updates; voting-based collision resolution and CMS minimum-value flow reconstruction; achieved **96% Top-K accuracy, 100% priority-packet recall**, and <1.1% throughput overhead at 10 Gbps on FABRIC testbed with CAIDA 2019 traffic.
- **High-Resolution eBPF System Resource Profiling for DML Workloads** | Paper / Code
  Developed a per-PID eBPF profiler with in-kernel Top-K tracking, delivering **10–14× finer temporal resolution** than top/htop, ~11 ms latency, and >90% Top-K fidelity on commodity Linux servers.
- **Scotch – Elastic SDN Overlay for Control-Plane Scalability** | Paper / Code
  Designed a distributed control-plane overlay using OvS-based traffic offloading and large-flow migration; **reduced controller congestion by >70%** on a 34-node GENI Clos topology.
- **Enterprise-Grade LLM Production Deployment Platform on Kubernetes** | Code
  Built a K8s microservices platform (Gateway / Router / Worker) supporting vLLM and TensorRT-LLM inference with ArgoCD GitOps, Prometheus/Grafana observability, and CI/CD automation across **50+ Kubernetes deployments**.
- **Domain-Adapted LLMs for Clinical Risk Adjustment Coding** | Paper
  Fine-tuned PubMedBERT and Qwen-7B for HCC coding from EHR notes; achieved **macro-/micro-F1 of 0.775/0.742** and **0.81/0.802** on MIMIC-IV with imbalance-aware training and per-label calibration.
- **BiLSTM-Based Side-Channel Modeling for Distributed ML Analysis** | Paper
  Designed a BiLSTM pipeline aligning CPU/memory energy traces with per-layer timestamps; reliably distinguishes Conv, FC, and activation layers during distributed ML training using ensemble learning.

RESEARCH & SYSTEMS ENGINEERING EXPERIENCE

Research Assistant, AI Infrastructure & ML Systems

Case Western Reserve University

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- **PRISM – Priority-Aware Transport for Distributed ML Training** | Paper (Under Review)
  Selective-reliability transport with magnitude-based gradient classification, dual congestion windows, and kernel-bypass (LibVMA); achieves **8–21% throughput gain over TCP** on 32 GPUs; matches NDP latency at 144-worker scale.
- **Training-Time Side-Channel Attacks on Distributed ML Systems** | Paper / Code
  Cross-correlated network traffic and Intel RAPL power side channels to reconstruct DNN architectures; achieved **>99% layer-type accuracy and <1.5% tensor-size error** — exposing MLaaS attack surfaces.
- **Priority-Based eBPF Network Flow Telemetry for AI / DML Infrastructure** | Paper / Code
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PUBLICATIONS

8 papers total, 7 first-author, 5 published at IEEE CCNC, IEEE ICNC, ACM TOIT, and Springer SecureComm; 3 under review (incl. ACM TOPS). Full list: johnny-dai-git.github.io/portfolio

PATENTS

CN 108594898 B — Oven Temperature Control Method, Device and Computer-Readable Storage Medium. Issued: July 23, 2021. Assignee: Midea Group Co., Ltd. Inventors: Shao Guangyuan, Dai Yuanjun, Zhu Jiele, Cheng Gang, Gong Jun.